KGiSL Institute of Technology

(Approved by AICTE, New Delhi; Affiliated to Anna University, Chennai)

Recognized by UGC, Accredited by NBA (IT)

365, KGiSL Campus, Thudiyalur Road, Saravanampatti, Coimbatore – 641035.

REGISTER NO : 711723205058

ROLL NO : 23IT58

NAME : B.THARASRI

DEPARTMENT : B.TECH.IT

SUBJECT : INDUSTRIAL SAFETY

SUBJECT CODE : MX3089

Toxic Gas Release Industrial Case Study

1. Introduction

The Bhopal Gas Tragedy is considered the deadliest industrial chemical disaster in history. It occurred on the night of 2–3 December 1984 in Bhopal at a pesticide plant operated by Union Carbide India Limited.

A highly toxic chemical, methyl isocyanate (MIC), leaked into the atmosphere, exposing hundreds of thousands of people living near the plant.

2. Background of the Industry

- The plant manufactured pesticides (mainly Sevin) using MIC as an intermediate chemical.
- MIC is extremely volatile, toxic, and reactive with water.
- Large quantities of MIC were stored in tanks for production efficiency.
- The plant was located close to densely populated residential areas.

3. Sequence of Events

Night of the Disaster

- Water entered MIC storage tank (Tank E610).
- MIC reacted violently with water (exothermic reaction).

- Temperature increased rapidly (above 200°C).
- Pressure built up inside the tank.
- Safety valves opened, releasing toxic gas into the atmosphere.
- Approximately 30–40 tons of gas leaked over 1–2 hours.

4. Flowchart of the Incident

Start
↓
Water enters MIC storage tank
↓
Chemical reaction begins (MIC + Water)
↓
Heat generation increases temperature
↓
Pressure builds up inside tank
↓
Safety valve opens
↓
Toxic gas (MIC) released
↓
Gas spreads over nearby areas
↓
Population exposed to toxic gas
↓
Mass casualties and environmental damage
↓
End

5. Causes of the Disaster

5.1 Technical Causes

- Entry of water into MIC tank due to poor isolation
- Lack of proper maintenance of pipelines and valves
- Corrosion in equipment
- Improper storage of large quantities of MIC

5.2 Safety System Failures

Multiple safety systems existed but were ineffective:

- Gas scrubber not functioning properly
- Flare tower was inoperative
- Refrigeration system was turned off to reduce cost
- Pressure and temperature monitoring systems were inadequate

5.3 Human and Organizational Causes

- Poorly trained workers
- Reduction in staff and safety personnel
- Ignoring earlier warning signs and minor leaks
- Lack of proper supervision
- Cost-cutting measures affecting safety operations

5.4 Management and Regulatory Failures

- Weak enforcement of industrial safety laws
- Lack of strict government monitoring
- Absence of emergency preparedness planning
- No proper communication system for nearby residents

6. Immediate Effects

6.1 Human Impact

- Thousands of people died within hours
- Severe symptoms observed:
  - Breathing difficulty
  - Eye irritation and blindness
  - Vomiting and chest pain
- Primary cause of death: lung damage (pulmonary edema)

6.2 Environmental Impact

- Toxic gas spread close to ground level
- Contamination of soil and water sources
- Death of animals and destruction of vegetation

7. Long-Term Effects

7.1 Health Effects

- Chronic respiratory diseases
- Eye damage and partial blindness
- Increased cancer rates
- Birth defects and genetic disorders in children

7.2 Environmental Effects

- Persistent groundwater contamination
- Hazardous waste left at the site
- Long-term ecological imbalance

7.3 Socio-Economic Effects

- Loss of livelihoods
- Increased poverty among affected families
- Inadequate compensation and rehabilitation

8. Root Cause Analysis

The disaster was not a single-point failure but a combination of:

- Technical failure: chemical reaction due to contamination
- Mechanical failure: non-functional safety systems
- Human error: poor training and negligence

9. Lessons Learned

9.1 Industrial Safety

- Hazardous chemicals must be stored in minimal quantities
- Continuous monitoring systems must be active
- Regular inspection and maintenance are essential

9.2 Process Safety

- Installation of multiple independent safety systems
- Automatic shutdown mechanisms
- Real-time monitoring of temperature and pressure

10. Conclusion

The Bhopal Gas Tragedy remains a critical example of industrial failure caused by negligence at multiple levels. It highlights the importance of safety in chemical industries and the consequences of ignoring it. The disaster led to global awareness and improvements in industrial safety standards, but it also serves as a warning that lack of responsibility in handling hazardous materials can lead to catastrophic human and environmental losses.